

Federated Machine Learning

Dataset Information

Cardiovascular Disease Dataset

- 70,000 instances
- 12 attributes

Binary Target Variable:

- Class 0 — No Cardiovascular Disease: 35,021
- Class 1 — Has Cardiovascular Disease: 34,979

Source:

<https://www.kaggle.com/datasets/salomemweluscherer/cardiovascular-disease-dataset>

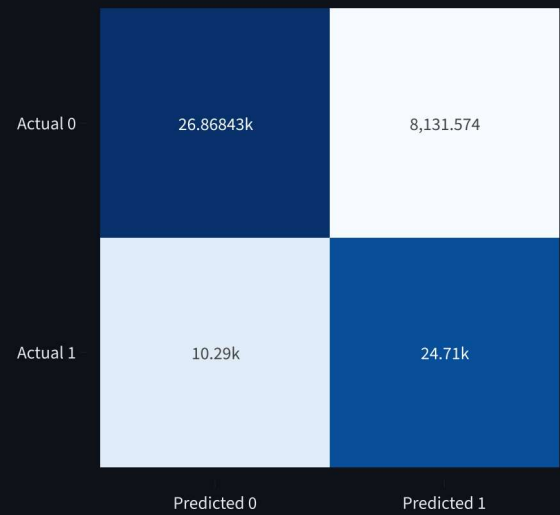
Run Federated Learning Pipeline

Pipeline Completed

Baseline Model Performance

	accuracy	precision	recall	f1	roc_auc
0	0.737	0.7524	0.706	0.7285	0.737

Confusion Matrix (Baseline Model)



Hospital Performance

Hospital 1 Analysis

Decision: retrained_due_to_reduction

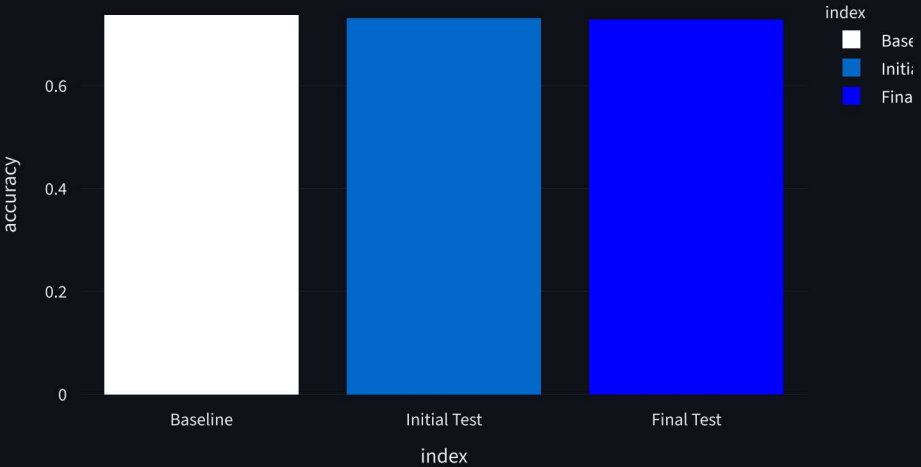
	accuracy	precision	recall	f1	roc_auc	samples
Baseline	0.737	0.7524	0.706	0.7285	0.737	None
Initial Test	0.7313	0.75	0.7057	0.7271	0.7317	1500
Final Test	0.728	0.7438	0.6951	0.7186	0.728	3000

Hospital 2 Analysis

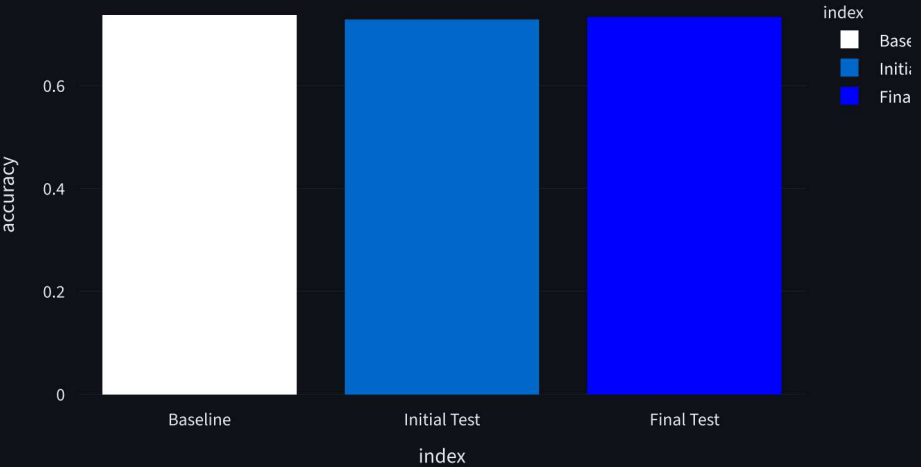
Decision: retrained_due_to_reduction

	accuracy	precision	recall	f1	roc_auc	samples
Baseline	0.737	0.7524	0.706	0.7285	0.737	None
Initial Test	0.7283	0.7518	0.6855	0.7171	0.7285	3000
Final Test	0.7332	0.7555	0.6893	0.7209	0.7332	4000

Hospital 1 Accuracy vs Baseline



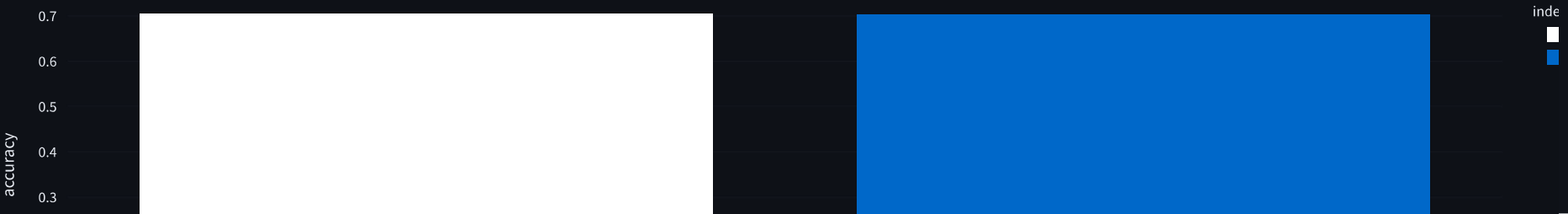
Hospital 2 Accuracy vs Baseline

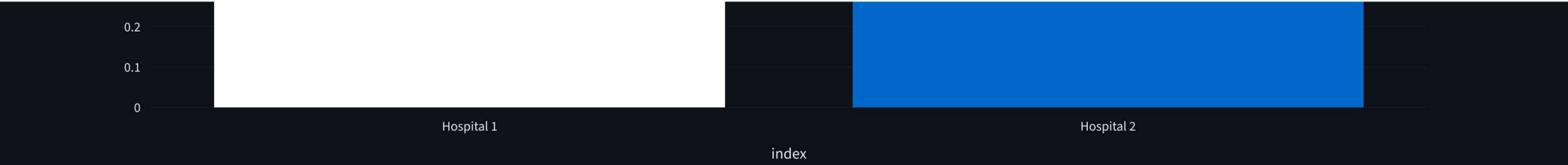


Global Model Performance (main_model_v2)

	accuracy	precision	recall	f1	roc_auc	samples
Hospital 1	0.705	0.7234	0.6631	0.692	0.705	3000
Hospital 2	0.703	0.7254	0.6528	0.6872	0.703	4000

Global Model Accuracy Across Hospitals





Accuracy Comparison

	Stage	Accuracy
0	Baseline	0.737
1	Hospital1 Initial	0.7313
2	Hospital1 Final	0.728
3	Hospital2 Initial	0.7283
4	Hospital2 Final	0.7332
5	Global Model (Hospital1)	0.705
6	Global Model (Hospital2)	0.703

Accuracy Comparison Across Federated Learning Stages



0

Baseline

Hospital1 Initial

Hospital1 Final

Hospital2 Initial

Hospital2 Final

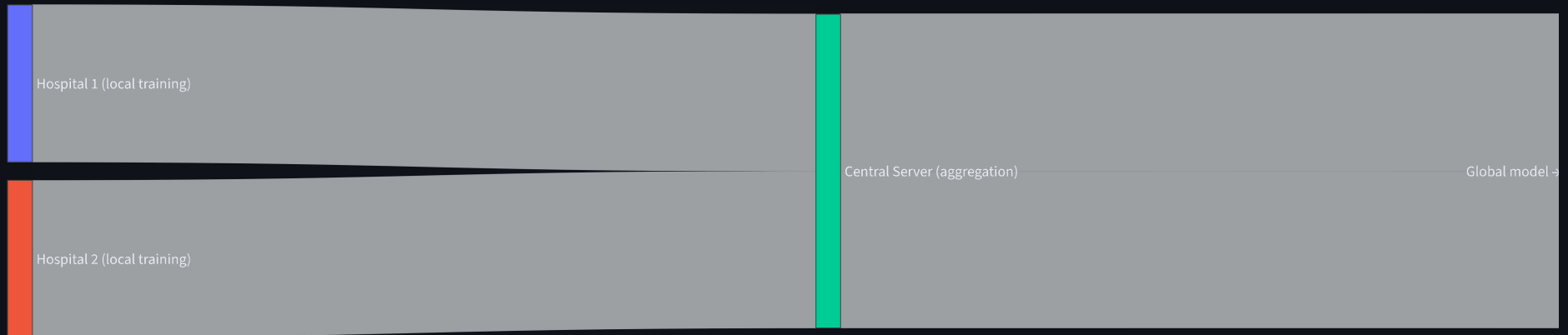
Global Model (Hospital1)

Global Model (Hospital2)

Stage

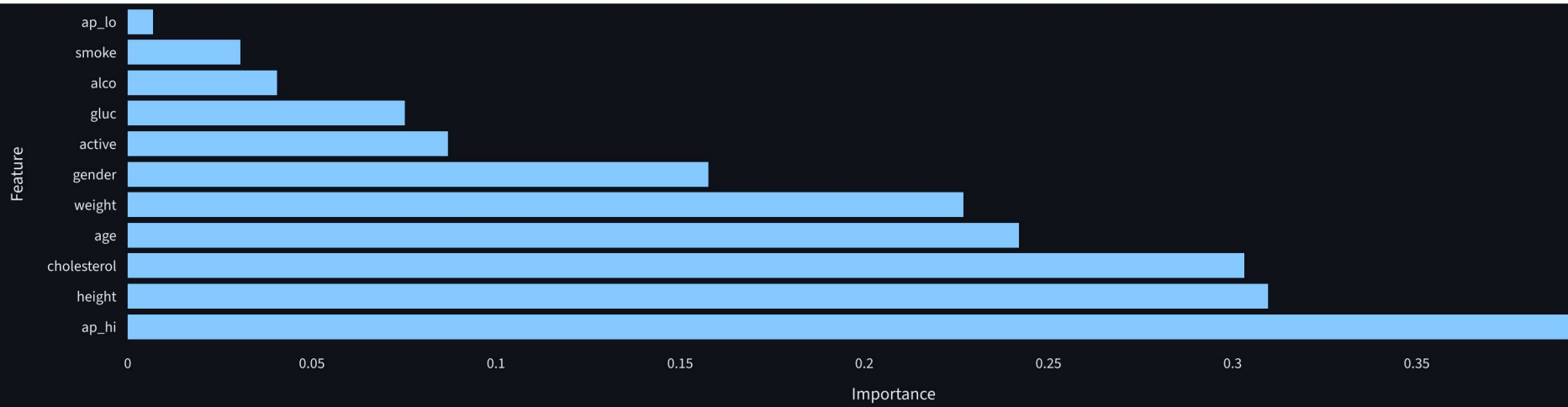
Federated Architecture (flow)

Federated Learning Data / Weights Flow (Sankey)



Explainability (SHAP)

Feature Importance for Cardiovascular Risk



	Feature	Importance
4	ap_hi	0.3951
2	height	0.3096
6	cholesterol	0.303
0	age	0.2419
3	weight	0.2269
1	gender	0.1576
10	active	0.0869
7	gluc	0.0751
9	alco	0.0405
8	smoke	0.0305